# Response to Review

**When does a spectral prior help graph learning?**
**Connectivity-loss estimation under road-network disruptions**

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We thank the reviewer for the careful and constructive assessment. We revised the manuscript to sharpen its network-science contribution, distinguish descriptive trends from inferentially supported comparisons, add transport robustness and size controls, and improve presentation. Locations below refer to the revised manuscript; source-line references are included to make the changes easy to audit before journal line numbering is applied.

## Major comments

**Reviewer comment.** *M1 — The contribution relative to residual-learning literature is not sufficiently explicit.*

**Response.** We agree. Section 2.3 now directly states that bounded residual learning itself is not claimed as novel and contrasts ResiliRoad with generic hybrid/physics-guided residual modelling. It identifies the network-science contribution as the controlled separation of (i) an explicit Fiedler perturbation prior, (ii) a Fiedler node feature, and (iii) a learned bounded finite-perturbation correction, followed by geographic transfer and correction calibration tests. We also added the closely related learnable-physics-engine literature and explain why the correction-to-prior calibration question is specific to spectral network disruption. See Section 2.3, p. 4 (source lines 134–159), and the bounded formulation in Section 3.

**Reviewer comment.** *M2 — Six country clusters provide limited inferential strength for leave-one-country-out analysis.*

**Response.** We now separate inference from description explicitly. The opening paragraph of the Discussion reports that four of the 12 primary expanded-OSM paired comparisons have clustered 95% intervals excluding zero and treats the remaining eight only as descriptive trends. The manuscript therefore does not claim population-level geographic generality from six countries. The Limitations section retains the small-cluster warning and states that larger pre-registered geographic samples are required. See Discussion and Limitations, p. 16 (source lines 604–643). The statistical unit and nested bootstrap are defined in Section 4.3 (source lines 333–354).

**Reviewer comment.** *M3 — The experiments omit established transportation-robustness comparators.*

**Response.** We added two non-spectral transport-topology controls: largest connected-component retention and origin–destination global-efficiency retention. Section 5.7 reports their association with the algebraic-connectivity-loss target and explains their intended role. They are useful controls for whether standard connectivity/reachability summaries explain the target, but are not presented as capacity- or demand-aware transport models. This avoids claiming that algebraic connectivity is a universally superior transport-service metric. See Section 5.7, p. 13 (source lines 489–509), with the full diagnostic in Appendix Figure 7.

**Reviewer comment.** *M4 — Synthetic-to-real transfer confounds graph size and street morphology.*

**Response.** We added a size-only control spanning 50–1,200 nodes within the same synthetic graph family. Section 5.7 now begins by stating that this is a within-family sensitivity analysis, not a morphology-matched transfer test. The result tests whether size alone reproduces the observed reversal, while the remaining inability to isolate morphology is retained as a limitation. See Section 5.7, p. 13 (source lines 489–509), and Limitations, p. 16.

## Requested final checks and presentation revisions

**Reviewer comment.** *Check the Journal of Complex Networks length and figure/table rules and reduce the runtime material if necessary.*

**Response.** We rechecked the current Instructions to Authors. They impose a 300-word abstract maximum and allow up to six keywords, but state no word-count or figure/table maximum for a Research Article. The revised manuscript has a sub-300-word abstract, six keywords, approximately 5,000 words by TeXcount, and seven figures and seven tables including the appendix. Nevertheless, we condensed Section 5.9 and retained only the runtime evidence needed to support the screening/scaling claim.

**Reviewer comment.** *Consolidate repeated statistical hedging.*

**Response.** Done. A single quantitative interpretive rule now opens the Discussion (p. 16): four of 12 primary expanded-OSM comparisons have clustered intervals excluding zero; the remaining eight are descriptive. Repetitive qualifiers were removed from the subsequent Discussion and Conclusion while the methods and result tables continue to report uncertainty transparently.

**Reviewer comment.** *Check float numbering and cross-references.*

**Response.** Done. Every figure and table is anchored at its first intended textual location, and all references use LaTeX `\label`/`\ref` pairs. The final build has no undefined references, duplicate labels, LaTeX errors, or overfull boxes. We also moved the architecture-fairness tables to the appendix, defined "biased anchor" at first use (Introduction, source lines 81–84), and increased axis-label sizes in the multi-panel diagnostics.

# Closing

No additional experiment is claimed to fully separate road morphology from other aspects of domain shift. We have instead bounded that conclusion and made the evidence hierarchy explicit. We believe the revised manuscript now states its novelty, empirical support, and limitations more precisely while remaining focused on its central network-science question.